

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416766
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Patel; Satish I.

Fusion spliced connector

Abstract

An optical connector which houses and holds a fusion spliced portion in which a short optical fiber fitted to an optical connector ferrule is fusion-spliced with a buffered optical fiber. One end of a protection sleeve which reinforces said fusion spliced portion is configured to be coupled to said optical connector ferrule. The protection sleeve has an inner sleeve and a clamping sleeve wherein the inner sleeve is configured to be inserted into the clamping sleeve. The end of the inner sleeve distal from the ferrule is configured to have a clamping feature that is activated when the inner sleeve is inserted into the clamping sleeve.

Inventors:	Patel; Satish I. (Roselle, IL)
Applicant:	Panduit Corp. (Tinley Park, IL)
Family ID:	1000008819840
Assignee:	Panduit Corp. (Tinley Park, IL)
Appl. No.:	17/978566
Filed:	November 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240142717 A1	May. 02, 2024

Publication Classification

Int. Cl.: G02B6/38 (20060101); G02B6/255 (20060101)

U.S. Cl.:

CPC G02B6/3846 (20130101); G02B6/2558 (20130101); G02B6/3851 (20130101);

Field of Classification Search

CPC: G02B (6/2551); G02B (6/2558); G02B (6/3846); G02B (6/3851); G02B (6/3888)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7011454	12/2005	Caveney	385/59	G02B 6/3879
7806600	12/2009	Marrs	385/60	G02B 6/3802
8047726	12/2010	Tamekuni et al.	N/A	N/A
8070367	12/2010	Winberg	385/75	G02B 6/38875
8317406	12/2011	Tamekuni et al.	N/A	N/A
8408811	12/2012	de Jong	385/69	G02B 6/3846
2009/0022457	12/2008	De Jong et al.	N/A	N/A
2020/0150374	12/2019	De Jong et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1800897	12/2005	CN	N/A
114488417	12/2021	CN	G02B 6/3855

OTHER PUBLICATIONS

English Translation for CN 114488417 A, 14 pages (Year: 2022). cited by examiner

Primary Examiner: Petkovsek; Daniel

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to fusion spliced fiber connectors and more specifically to a method and apparatus of protecting the fusion spliced fiber in the connector.

BACKGROUND

(2) To make a connection with field fiber using a splice-on connector approach, Panduit currently offers mechanical splice-on connectors under Opticam name in LC, SC and ST styles as detailed in U.S. Pat. Nos. 7,011,454 and 8,256,970. The Opticam connectors feature factory polished ferrule with stub fiber disposed in index matching gel inside the connector. For termination, field fiber is stripped, cleaned, and cleaved to a set length and inserted into the connector to align with the stub fiber through index matching gel. Once field fiber is aligned, cam action buffer clamp secures the field fiber into place to prevent any movement. The Opticam mechanical splice-on connectors are easy to terminate and do not require expensive termination tools. It uses index matching gel to transfer signal from field fiber to stub fiber. As optical property of the glass and gel is not identical, mechanical splice-on connectors have higher insertion loss.

(3) What is needed is a fusion splice type splice-on connector where glass is fused thermally to offer lowest loss connector. This type of connector was not preferred in the industry due to the extremely high cost of the fusion splice equipment, but lately cost of fusion splice equipment has

come down to make this type of connector cost effective for field use.

SUMMARY

(4) An optical connector which houses and holds a fusion spliced portion in which a short optical fiber fitted to an optical connector ferrule is fusion-spliced with a buffered optical fiber. One end of a protection sleeve which reinforces said fusion spliced portion is configured to be coupled to said optical connector ferrule. The protection sleeve has an inner sleeve and a clamping sleeve wherein the inner sleeve is configured to be inserted into the clamping sleeve. The end of the inner sleeve distal from the ferrule is configured to have a clamping feature that is activated when the inner sleeve is inserted into the clamping sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows trimetric view of fusion splice connector assembly **10**.
- (2) FIG. 2 is an exploded view of fusion splice connector **20**.
- (3) FIG. 3 is an exploded 180-degree rotated vertical axis view showing protective sleeve **100** cable buffer clamping end detail “A” and ferrule assembly **60** detail “B” locations.
- (4) FIG. 4 is protective sleeve **100** cable buffer clamping end detail “A” view scaled to show buffer clamp **150**.
- (5) FIG. 5 is a detail “B” view scaled to show ferrule assembly top keying notch **170**, side keying notch **180** and sleeve retaining feature **160**.
- (6) FIG. 6 is an exploded trimetric view rotated 180-degrees vertical axis and axis passing thru the connector.
- (7) FIG. 7 is a detail “C” view showing ferrule holder flat **190** on ferrule holder **80**.
- (8) FIG. 8 is a trimetric view of protective sleeve **100** showing inner keying flat **200**, retaining groove **210**, outer keying flat **220** and buffer clamp **150**.
- (9) FIG. 9 is a trimetric view of clamping sleeve **110** showing clamping sleeve keying flat **230** and tapered section **240**.
- (10) FIG. 10 is a trimetric view of connector housing **50** showing ferrule top keying feature **290** and ferrule side keying feature **300**.
- (11) FIG. 11 is an exploded view showing assembly step **1**.
- (12) FIG. 12 shows assembly step **2**.
- (13) FIG. 13 shows assembly step **3**.
- (14) FIG. 14 shows assembly step **4**.
- (15) FIG. 15 shows assembly step **5**.
- (16) FIG. 16 shows assembly step **6**.
- (17) FIG. 17 shows assembly step **7**.
- (18) FIG. 18 shows assembled connector **10**.
- (19) FIG. 19 is a top view showing section lines **19** and **21**.
- (20) FIG. 20 is a section view passing thru line **19** in FIG. 19.
- (21) FIG. 21 is a detail “D” showing clamping sleeve taper section **240** pressing buffer clamps **150** in a clamping position over cable **30**.
- (22) FIG. 22 is a section view along line **21** in FIG. 19.

DESCRIPTION OF THE INVENTION

(23) The present invention is a splice-on fusion splice connector with a splice protection sleeve featuring mechanical buffer clamps. The invention shown here is for an LC style connector, but this invention can be implemented in SC or ST style fiber optic connectors.

(24) FIG. 1 shows trimetric view of fusion splice connector assembly **10**. Fusion splice connector **20** is terminated to fiber optic cable **30**.

(25) FIG. 2 is an exploded view of fusion splice connector **20**. Fusion splice connector **20** includes ferrule dust cap **40**, connector housing **50**, ferrule assembly **60**, protective inner sleeve **100**, clamping sleeve **110**, spring **120**, backbone **130** and boot **140**. Ferrule assembly **60** includes ferrule **70**, ferrule holder **80** and fiber **90**.

(26) FIG. 3 is an exploded 180-degree rotated vertical axis view showing protective sleeve **100** cable buffer clamping end detail “A” and ferrule assembly **60** detail “B” locations.

(27) FIG. 4 is protective inner sleeve **100** cable buffer clamping end detail “A” view scaled to show buffer clamp **150**.

(28) FIG. 5 is a detail “B” view scaled to show ferrule assembly top keying notch **170**, side keying notch **180** and sleeve retaining feature **160**.

(29) FIG. 6 is an exploded trimetric view rotated 180-degrees vertical axis and axis passing thru the connector. Detail “C” is called out in this view.

(30) FIG. 7 is a detail “C” view showing ferrule holder flat **190** on ferrule holder **80**.

(31) FIG. 8 is a trimetric view of protective inner sleeve **100** showing inner keying flat **200**, retaining groove **210**, outer keying flat **220** and buffer clamp **150**. During assembly ferrule holder flat **190** is aligned with the protective sleeve inner keying flat **200** to prevent protective sleeve rotation and retaining groove **210** slides over ferrule holder sleeve retaining feature **160** to prevent it from coming off.

(32) FIG. 9 is a trimetric view of clamping sleeve **110** showing clamping sleeve keying flat **230** and tapered section **240**. Protective sleeve outer keying flat **220** aligns with clamping sleeve keying flat **230** to prevent clamping sleeve **110** rotation. Clamping sleeve tapered section **240** forces protective inner sleeve **100** buffer clamps **150** towards the fiber cable **30** to create a firm clamping force.

(33) FIG. 10 is a trimetric view of connector housing **50** showing ferrule top keying feature **290** and ferrule side keying feature **300**.

(34) FIG. 11 is an exploded view showing assembly step 1. In this step boot **140**, backbone **130**, spring **120**, clamping sleeve **110** and protective sleeve **100** are threaded on to fiber optic cable **30**.

(35) FIG. 12 shows assembly step 2. In this step buffer is stripped, and glass fiber **250** is cleaved to a specified length.

(36) FIG. 13 shows assembly step 3. In this step ferrule assembly **60** assembled with ferrule dust cap **40** is fusion spliced to cleaved glass fiber **250** at fusion splice point **260** using fusion splice equipment.

(37) FIG. 14 shows assembly step 4. In this step protective sleeve is slipped over the ferrule assembly while aligning protection sleeve inner keying flat **200** with ferrule holder flat **190**. Protective sleeve is pushed all the way to engage sleeve retaining feature **160** with retaining groove **210**.

(38) FIG. 15 shows assembly step 5. In this step clamping sleeve **110** is slipped over the protective sleeve **100** while aligning protective sleeve outer keying flat **220** with clamping sleeve keying flat **230**. Clamping sleeve tapered section **240** pushes buffer clamp **150** to clamp the cable **30**.

(39) FIG. 16 shows assembly step 6. In this step ferrule dust cap **40** is removed, spring **120** and backbone **130** are positioned to assembled with the housing **50** to engage backbone latches **280** in housing latch windows **270**. Housing **50** ferrule top keying feature **290** and ferrule side keying feature **300** are aligned to top keying notch **170** and side keying notch **180** respectively.

(40) FIG. 17 shows assembly step 7. In this step boot **140** is pushed over the backbone **130** to complete the assembly.

(41) FIG. 18 shows assembled connector **10**.

(42) FIG. 19 is a top view showing section lines **19** and **21**.

(43) FIG. 20 is a section view passing thru line **19** in FIG. 19.

(44) FIG. 21 is a detail “D” showing clamping sleeve taper section **240** pressing buffer clamps **150** in a clamping position over cable **30**.

(45) FIG. 22 is a section view along line 21 in FIG. 19. It shows crimp sleeve tapered section 240 pressing on buffer clamps 150.

Claims

1. An optical connector which houses and holds a fusion spliced portion where a short optical fiber fitted to an optical connector ferrule is fusion-spliced with a buffered optical fiber comprising: a housing; said fusion spliced portion; and a protection sleeve within the housing which reinforces said fusion spliced portion and is configured to be coupled to a ferrule holder, wherein said protection sleeve includes an inner sleeve and a clamping sleeve wherein the inner sleeve is configured to be inserted and fully enclosed by the clamping sleeve and further wherein an end of the inner sleeve distal from the ferrule holder is configured to have a clamping feature that is activated when the inner sleeve is inserted into the clamping sleeve.
 2. The optical connector of claim 1, wherein the clamping feature is a buffer clamp on the inner sleeve that is activated by a tapered section of the clamping sleeve engaging the buffer clamp of the inner sleeve.
 3. The optical connector of claim 1, wherein the ferrule holder has a sleeve retaining feature that is configured to engage a retaining groove on the inner sleeve.
 4. The optical connector of claim 1, wherein the ferrule holder has a ferrule holder flat configured to engage an inner keying flat on the inner sleeve.
 5. The optical connector of claim 4, wherein the inner sleeve also has an outer keying flat configured to engage a keying flat of the clamping sleeve.
-